

# Volume II · Chapter 10 — Benchmarking & Hallucination · Labs

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-II-Chapter-10-context.md for the AI interaction log.

**Prerequisites:** Ch 8, Ch 9.

**Learning outcomes — the student can:** evaluate LLMs on medical benchmarks (MedQA/PubMedQA/MedMCQA); interpret results; measure hallucination; build an eval harness; explain why benchmark ≠ clinical safety.

**Labs (hands-on) — 8 h:**

| #   | Lab                                                           | h |
|-----|---------------------------------------------------------------|---|
| 10a | Run a model on a MedQA-style set; compute accuracy            | 3 |
| 10b | Build a small eval harness with the evaluate library          | 2 |
| 10c | Hallucination test: craft adversarial clinical prompts, score | 3 |

**Datasets/tools:** HF datasets/evaluate; local LLM. **Assessment:** eval-harness deliverable (**50%**); hallucination report (**30%**); quiz (**20%**). **Key decisions:** which benchmark supports which claim; honest reporting of limitations. **References:** plan §13 → *Fine-tuning, alignment & benchmarks; Medical LLMs*. **Hours:** Theory 8 + Lab 8 = 16.